Research project 2

Investigating video preprocessing for Russian Sign Language translation:

quality constraints and computational efficiency

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Contents

1. A few words on sign language translation

2. Problem-setting and related work

3. Theoretical background

4. Research question and goals

5. Project timeline

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6. Study design:
data collection and preparation, model architecture

7. Experiment setup

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8. Model evaluation and comparison

9. Conclusions and limitations

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Sign language translation

What we take for granted

Sign languages…

- are natural languages
- are not mutually intelligible
- have their own grammar
- allow to express complex abstract meanings
- belong in general linguistic theory

What’s at stake

Linguistically (Kibrik, 2012):

- (re-)constructing the notion of natural human language by determining modality-independent *universals* and auditory-vocal or visual-gestural *variation*

Socially (Kibrik, 2012):

- *striving* for language equality – access to opportunities and information, social participation
- *encountering* a gap between the number of speakers and the number of teachers and interpreters – 9 times less than needed for RSL (Russian census, 2010)

Technologically (Fink et al., 2023):

- developing and deploying tools, not only for sign language translation into written text, but also Sign2Sign translation, chatbots, etc. and for real-time use!

What it takes

1. More *usable* data, especially for *continuous natural* speech (existing datasets suffer from limited domain coverage, annotation alignment inconsistencies and lack of signer diversity)

(Toshpulatov et. al, 2025)

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2. More computing power

(Toshpulatov et. al, 2025)

What it takes

1. More *usable* data, especially for *continuous natural* speech (existing datasets suffer from limited domain coverage, annotation alignment inconsistencies and lack of signer diversity)

2. ~~More computing power?~~
   Smart resource use, both in training and deployment, especially for real-time applications

(Toshpulatov et. al, 2025)

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What’s on the table

- Selecting most informative video frames (Damdoo, Kumar, 2025)
- Cropping most informative frame segments (Zhou et al., 2022)
- Extracting pose information (Orbay, Akarun, 2020)
- Sign2Gloss2Text pipelines (Maia et al., 2025)*
- Using unlabeled data or combining datasets
- More compact architectures
- etc. etc.

* a **gloss** is a written or typed approximation of (or notes regarding) each sign, a word-like label

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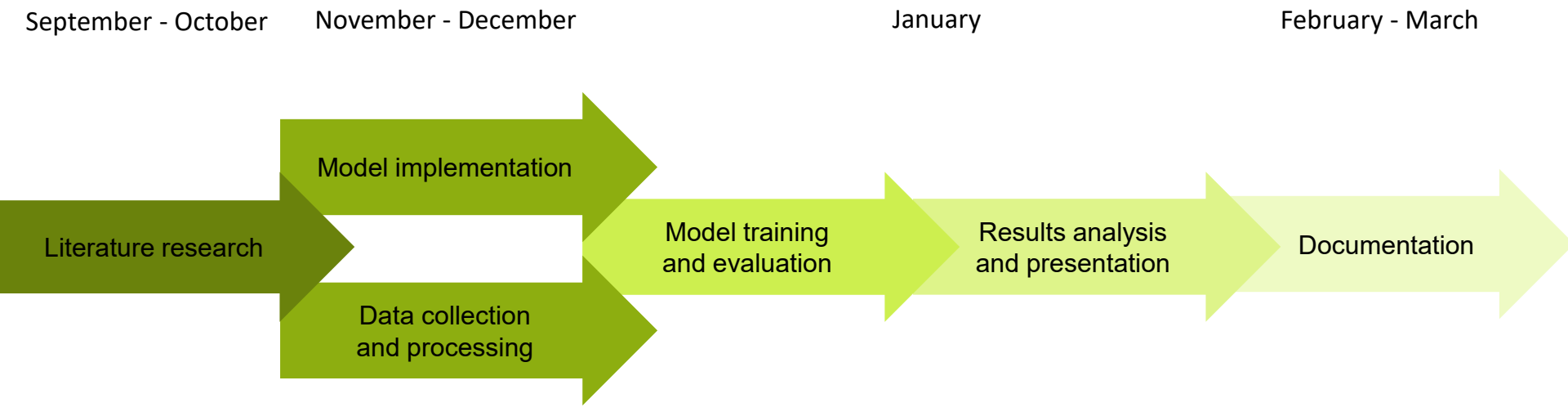
Project timeline



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Data collection and preparation

Dataset – ?

“Slovo” (Kapitanov et al., 2023) and “Bukva” (Kvanchiani et al., 2024) – isolated signs

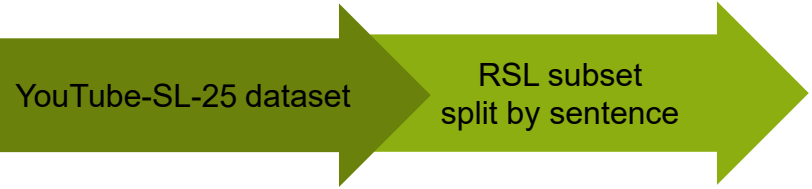
Russian Sign Language corpus (Novosibirsk State Technical University, 2014 -)

- fully-glossed and translated spontaneous monologues and dialogues
- 45 deaf, hard-of-hearing, deaf-blind and CODA signers, 18-63 years old
- 240 files / 5,5 hours of video total
- limited access

Dataset – ?

“YouTube-SL-25” (Tanzer and Zhang, 2024)

- a large-scale, open-domain corpus of sign language videos from YouTube
- more than 300 sign languages total: 715 videos / 60 hours represent RSL
- seemingly well-aligned captions (verified manually at random)
- signer diversity and topic variety – ?
- data quality – ?



715 => 710 videos with captions fully in Russian:

- 9 different channels (possibly, more individual signers)
- several common topics based on video category and tags
- highly varied video duration potentially skews the distribution of topics

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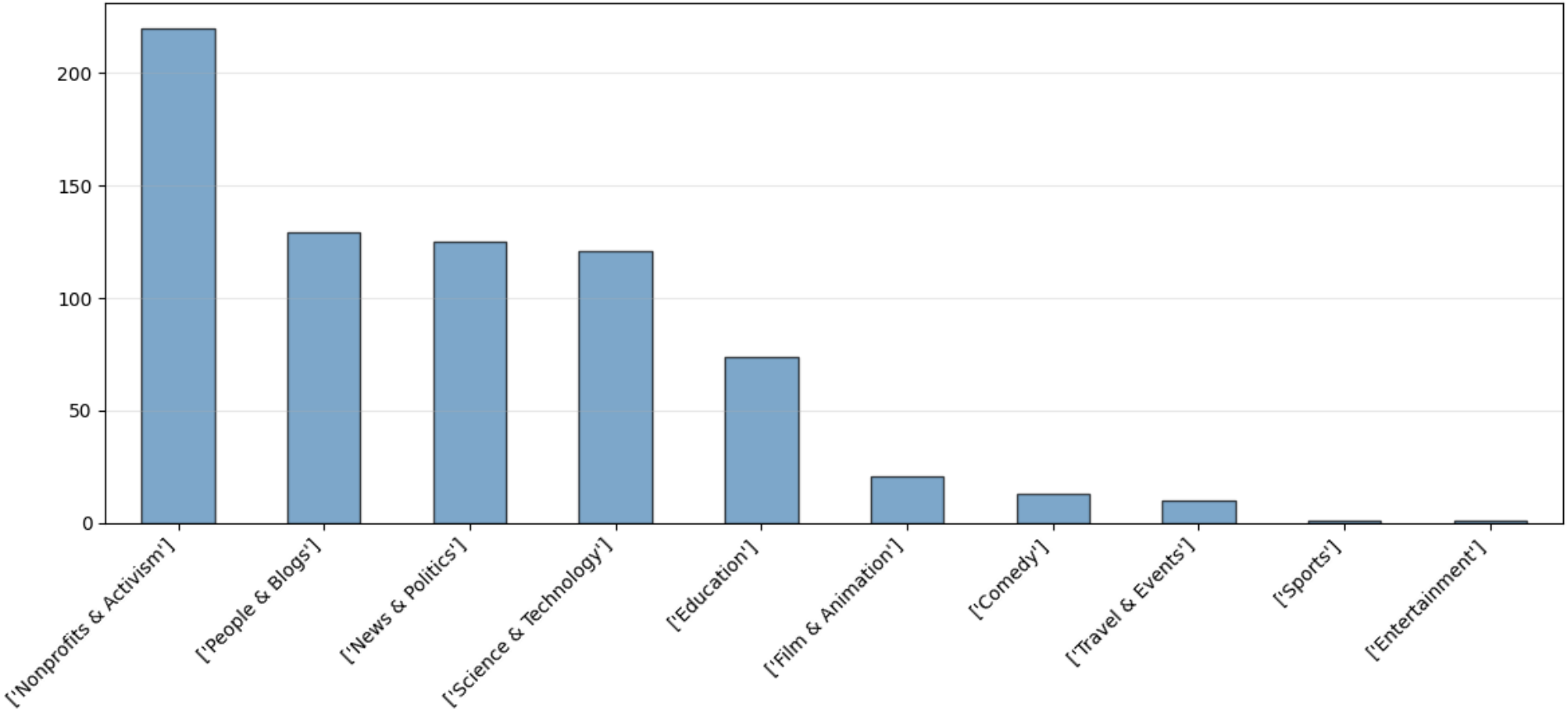

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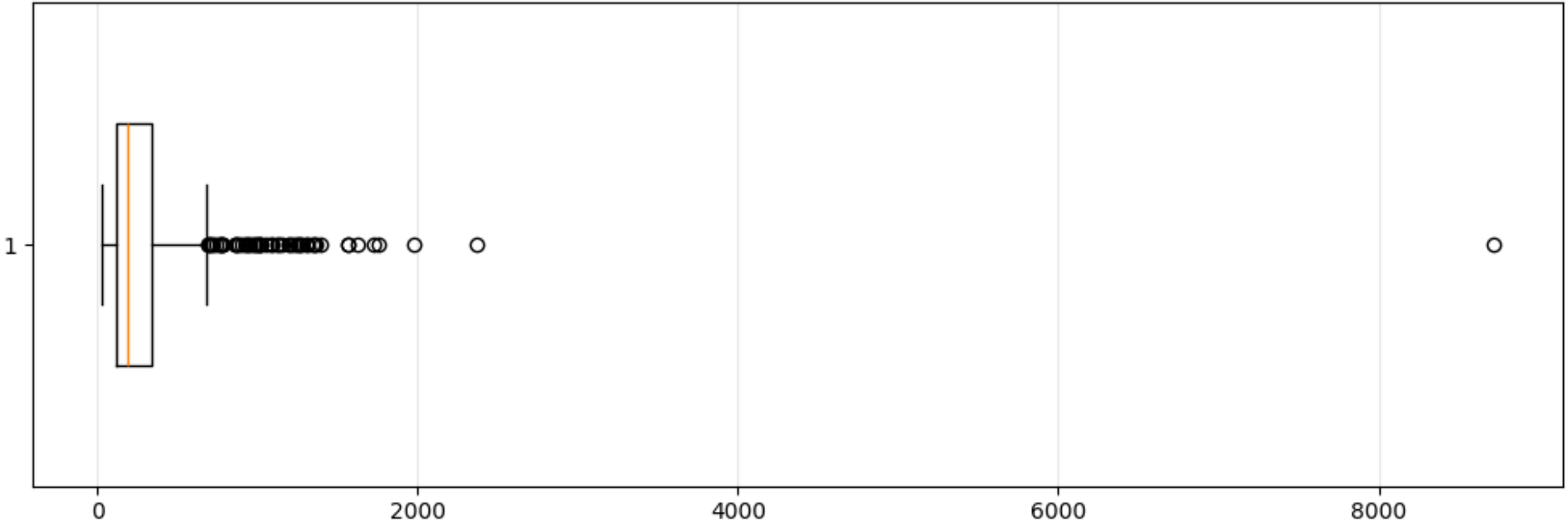
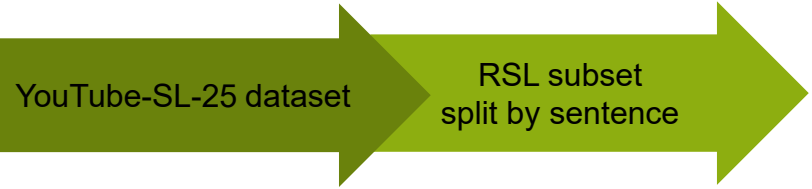
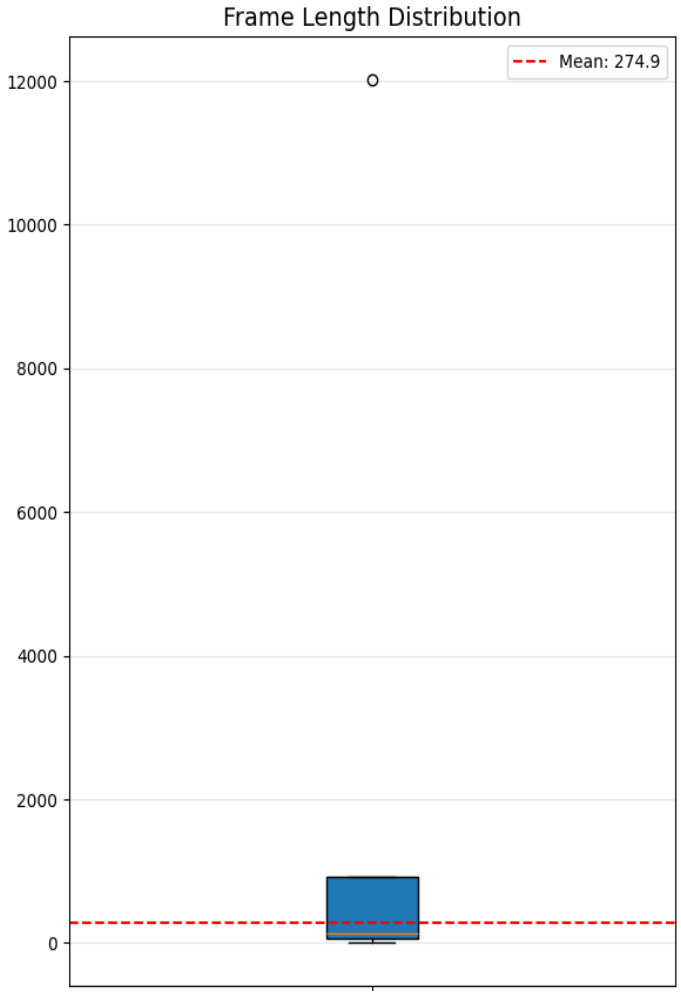
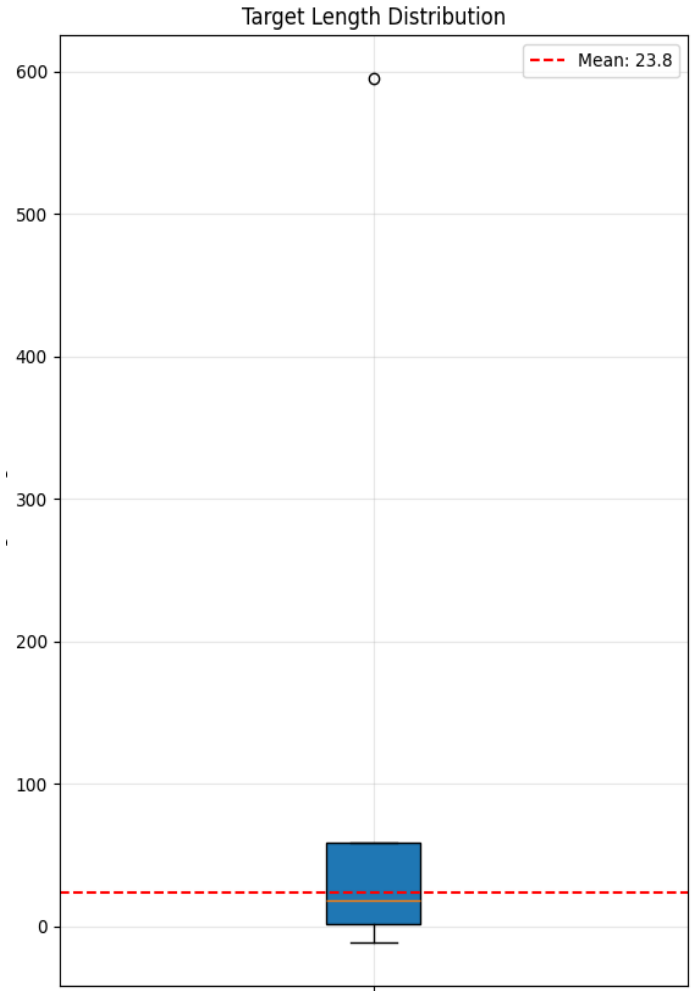

1
0
2000
4000
6000
8000

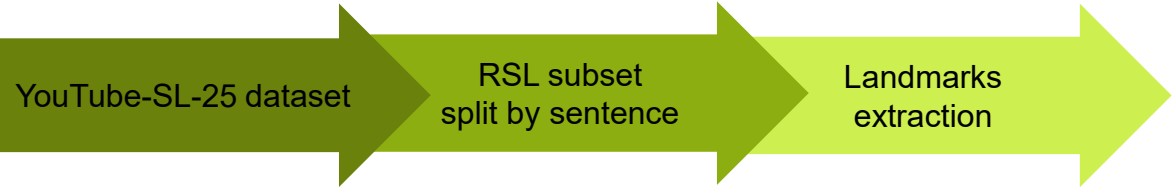


715 => 710 videos => roughly 18 000 *sentences*

- sentence boundaries based on captions
- caption segments combined until end-of-sentence punctuation
- several short sentences in a single caption segment remain due to the lack of information about the actual sentence boundary
- unreasonably long outliers point out messed up captions
- videos cut based on sentence boundaries


Frame Length Distribution
Mean: 274.9
12000
10000
8000
6000
4000
2000
0


Target Length Distribution
Mean: 23.8
600
500
400
300
200
100
0



How do we turn video input in a comprehensive representation of sign language?

Linguistic theory:

Signs are not fixed, but constructed from simultaneous components – *designator* hand shape, *tabula* location and *signation* movement or action (Stokoe, 1960)

Both a system of notation and a way of breaking the meaning into “morphemes”

Neither of the components is represented by a fixed set of options

=> We need a way to encode these main components

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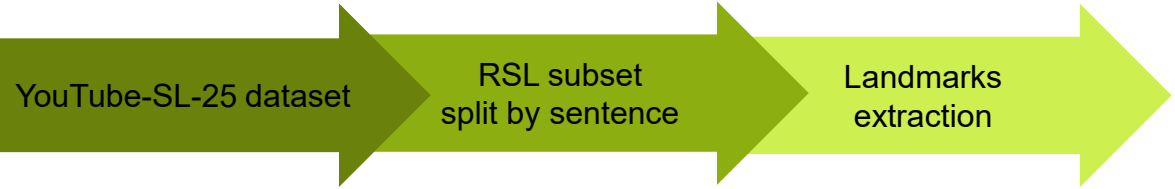

=> We need a way to encode these main components

=> MediaPipe framework (Lugaresi et al., 2019):

- minimal memory transfer between inference backends during processing
  => ideal for real-time applications (Lugaresi et al. 2019)
- shown to outperform OpenPose on angle measurement with comparable detection accuracy (Nguyen et. al, 2023)
- now separate tasks and models for capturing pose and hand movements

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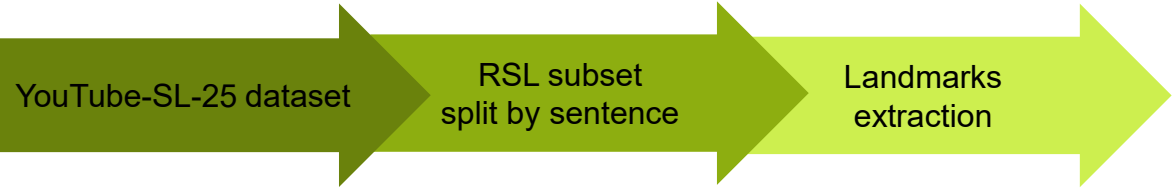


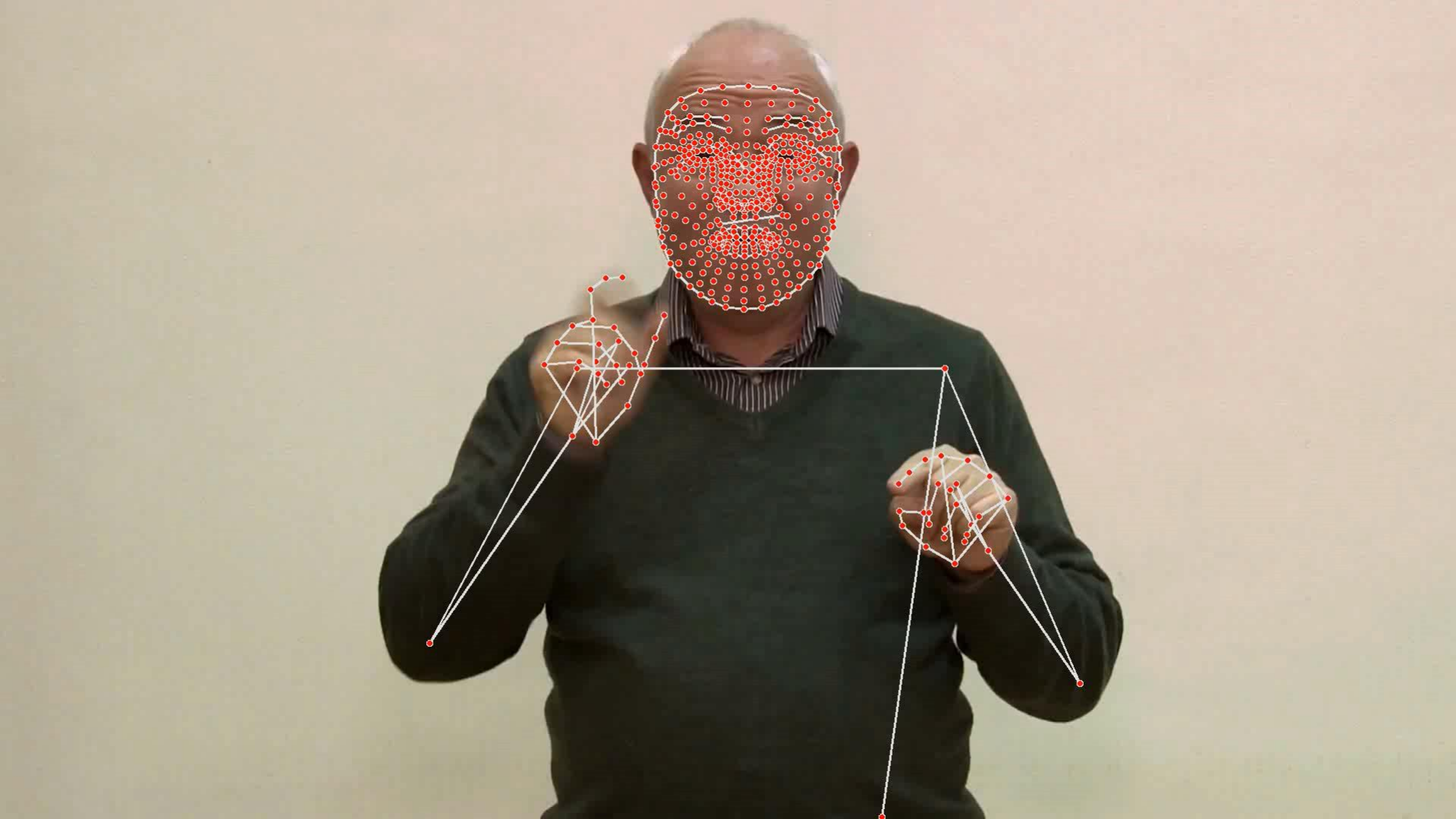
How do we turn video input in a comprehensive representation of sign language?

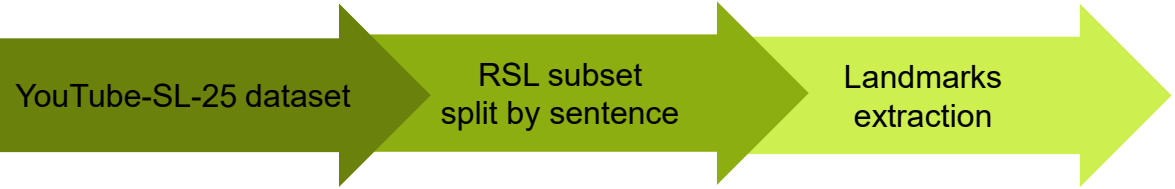
Linguistic theory:

Non-manual components are not exclusively emotional (Kimmelman, 2020) and follow language rules (Baker-Shenk, 1985)

They play a significant role in phonology, prosody, morphology, syntax, pragmatics (Pfau, 2010)

In longer texts specifically, their communicative value is so high, that during a dialogue the interlocutors focus more on each other's face, than hands (Siple, 1978)





How do we turn video input in a comprehensive representation of sign language?

MediaPipe framework:

- Pose + hands = 59 keypoints
- Face = 478 keypoints => 537 keypoints total

While deemed very efficient, MediaPipe took about 20 hours processing my dataset with 24 threads!

Stepping back: a research question!

9 times more values to process per sample
=> higher demand on memory and computing power, longer training time

In exchange: highly valuable information – from a language stand-point!

Is this information valuable for automatic sign language translation?

Does the value outweigh the expense?

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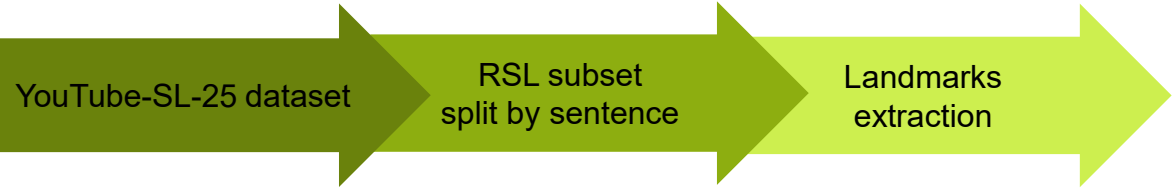

715 => 710 videos => roughly 18 000 sentences => 15609 sentences with detected motion

- sentence boundaries are inferred from captions and videos are cut based on these boundaries
- padding experiment on a subset has not yielded a significant improvement
- filter removes both cases with mistimed boundaries and non-signing content (i.e. credits)
- exceptions management:
  MediaPipe execution completely crashed on some videos, some videos could not be read properly, etc.

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… closely follows the Maia et al. (2025) approach to introduce more variability to the data and thus increase the generalization capabilities of the model. The process consists of:

- scaling by a random factor (mimics camera distance and body size variation)
- shearing along a random axis (mimics lens distortion or view from aside)
- rotation by a random angle (mimics camera tilt or non-frontal view)
- shifting by a random vector (mimics the signer standing off-center)

=> normalizing the data to have zero mean and unit variance

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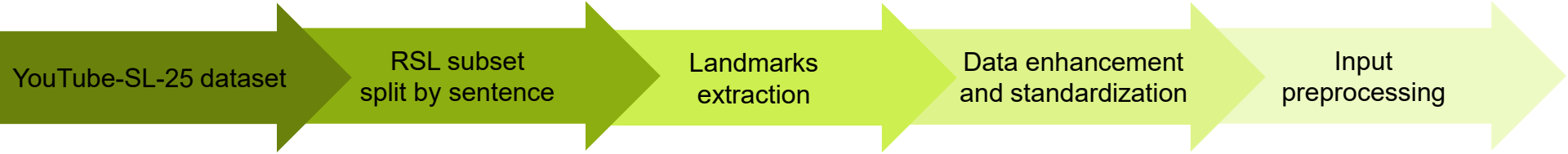

Finally, a few standard steps!

- target captions are tokenized with the ruBERT tokenizer (Kuratov, Arkhipov, 2019), converted to an integer sequence and truncated or padded to max. 64 tokens
- input keypoints are truncated or padded to max. 160 frames

=> single input sample shape (160, 177) vs. (160, 1611)
=59x3 =537x3

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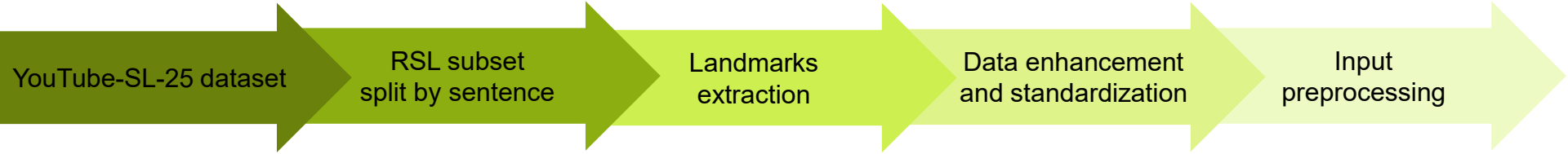

Finally, a few standard steps!

- 80-10-10 data split
- sentences from each original video kept together
- stratification on channel id

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Model architecture and training

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Model architecture

(Maia et al., 2025)

First model trained on MediaPipe body keypoints (hands + pose) to achieve translation quality comparable to existing baseline models

- Pipeline with intermediate step: Sign2Gloss2Text
- High-quality curated datasets of short sentences
- Almost 40 000 data samples
- Aggressive data augmentation

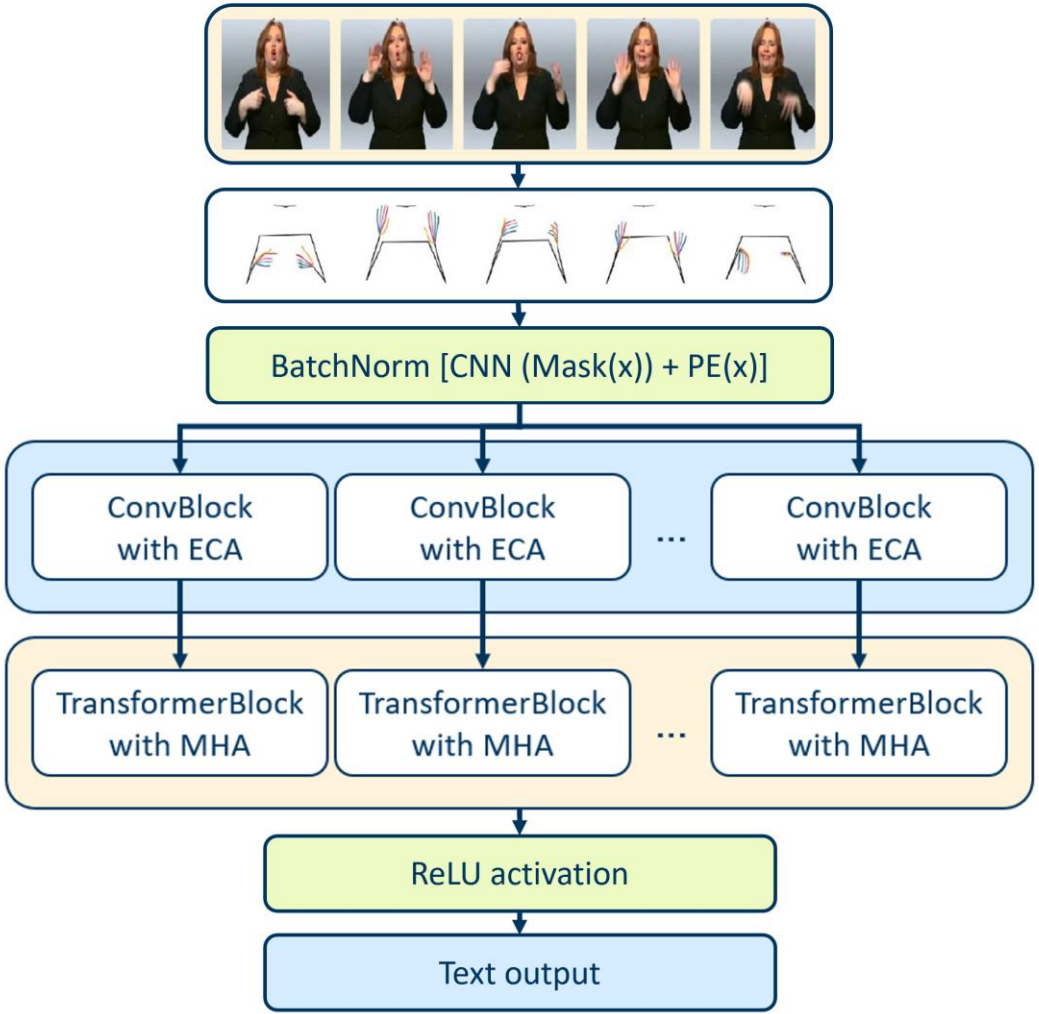
Stepping back: another research question!

Working with body keypoints and the same extraction framework

However: semi-annotated data, inconsistent filming, etc.

Does more realistic input data introduce too much noise for the model to learn well?

Or does the realistic noise essentially cover what data augmentation does?



(Maia et al., 2025)

Model architecture

- Same layers, blocks, functions => optimization through width and depth
- Same optimizer (AdamW) and loss* function => optimization through hyperparameters
- Input frame length and target length – adjusted based on data
- Batch size – adjusted based on available RAM

* **C**onnectionist **T**emporal **C**lassification loss for sequence-to-sequence tasks with unknown alignment between input and output (Graves et al., 2006):

$$CTCLoss(\overline{y}, y) = -\sum_{(y,\overline{y})} \log p(y|\overline{y})$$

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Experiment setup

Two main training configurations:

- only hands + pose keypoints
- hands + pose + face keypoints

For a fair comparison, hyperparameters are optimized to match the size and the complexity of the data

Additional optimization concern: data augmentation on train set
=> two more training runs under the same conditions

All trained on an NVIDIA H100 80GB HBM3 GPU for max. 100 epochs with early stopping (patience = 10)

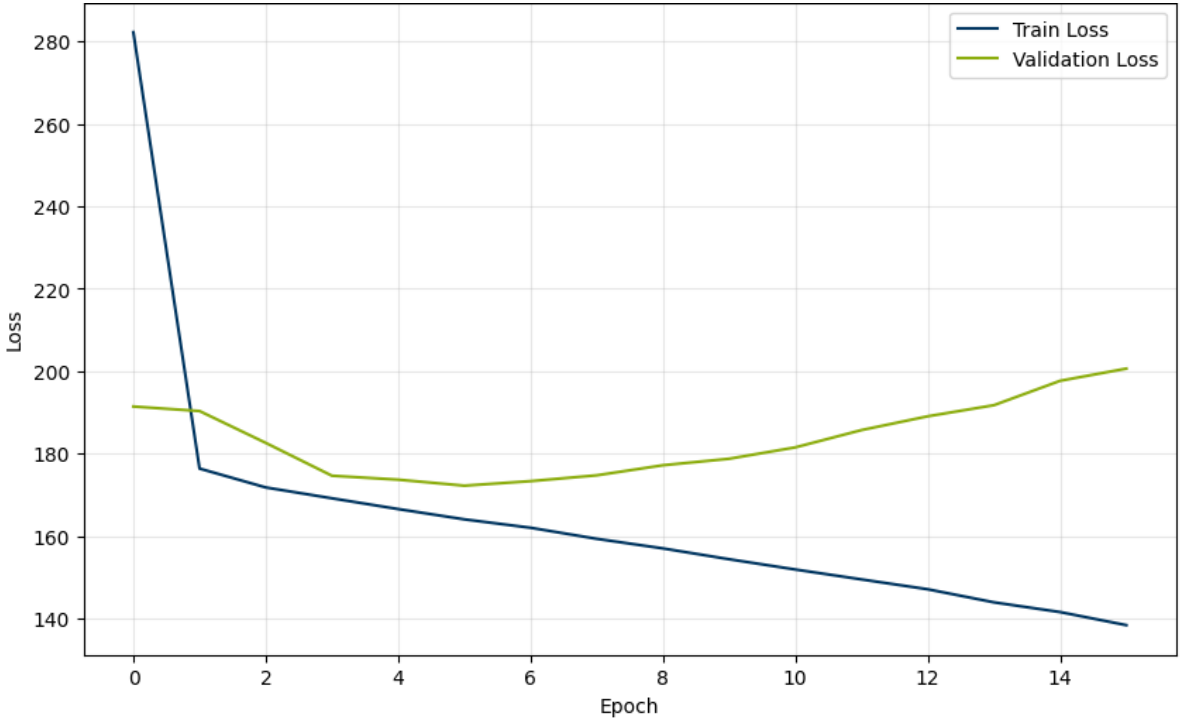
Training and validation loss are tracked per epoch with model check points based on validation loss

Hyperparameters

	Maia et al. 2025	hands + pose	+ face
Transformer blocks	6	2	4
Transformer dim	384	256	256
Dropout rate	0.6	0.4	0.5
Learning rate	1×10^{-4}	1×10^{-4}	1×10^{-4}
Weight decay	1×10^{-4}	1×10^{-3}	1×10^{-3}

Model evaluation and comparison

Training summary: hands + pose



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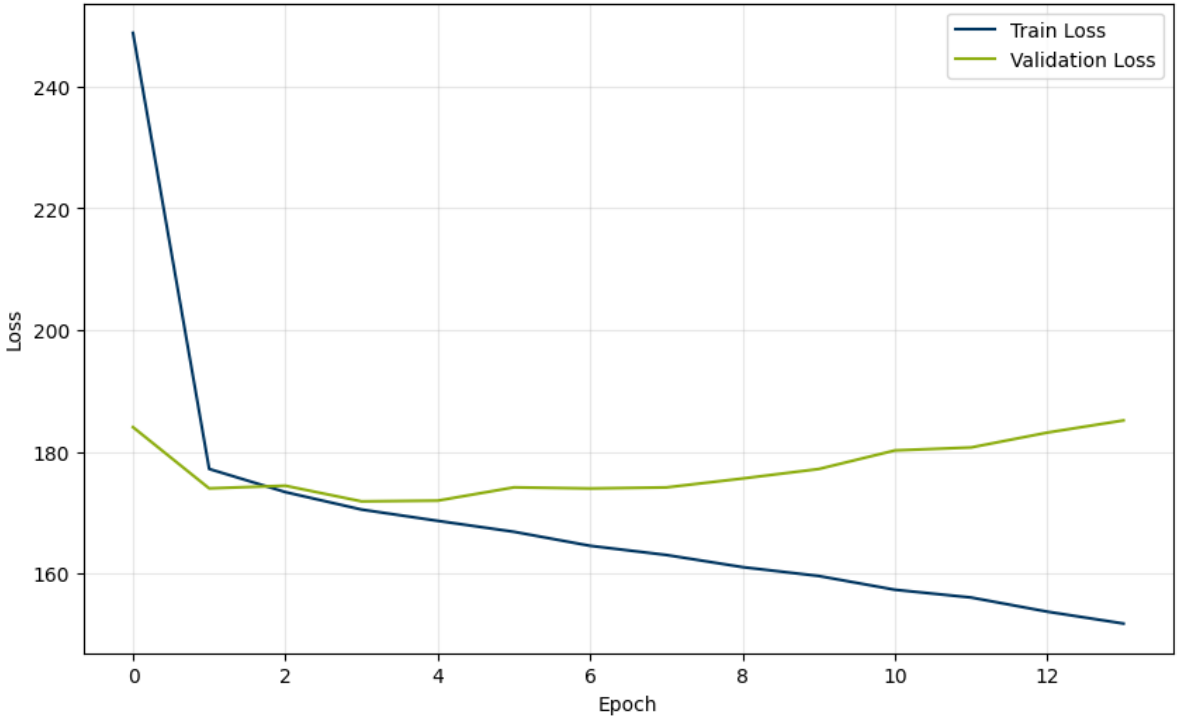
Training summary: with face, raw



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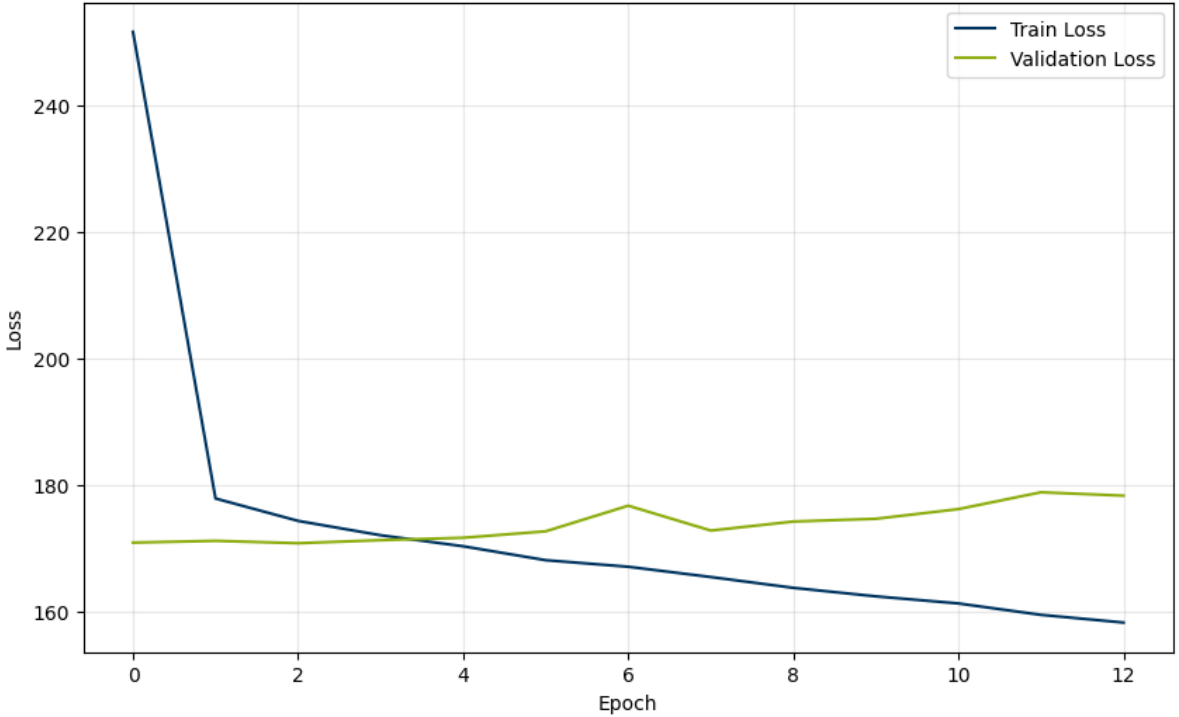
Training summary: with face, augmented



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Evaluation of models

	hands + pose	+ face
Trainable parametres	107 451 831	108 021 687
Total training time	18 min.	20 min.
Peak GPU memory	10 970.01 MB	11 018.80 MB
FLOPs	20 439 132 864	21 333 620 032
CO2 emissions	0.0330 kg	0.0347 kg

=> negligible difference in computational demand
=> focus on better performance!

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Evaluation of models

1. Translation quality: BLEU score

Evaluation of models

0. Random output test

1. Translation quality: BLEU score

Evaluation of models

0. Random output test

[UNK] [UNK] [UNK] я [UNK] [UNK] в [UNK]

Despite verified tokenization on input data!

1. ~~Translation quality: BLEU score~~

Evaluation of models

0. Random output test

[UNK] [UNK] [UNK] я [UNK] [UNK] в [UNK]

Despite verified tokenization on input data!

1. ~~Translation quality: BLEU score~~

2. ~~Translation quality: is data augmentation justified?~~

3. ~~Translation quality: is the inclusion of facial keypoints justified?~~

Results and conclusions

Why it failed? Too much noise in the data?

- multiple signers in one video
- busy backgrounds (incl. other people)
- poor sentence boundaries (incl. cutting signs) due to imperfect video-caption alignment
- inconsistent captions (incl. typos)
- inconsistent camera angles (-> keypoints relations) and video quality (-> keypoints detection)

Bottom line: unfavorable cost-benefit trade-off

Results and conclusions

What have we learned?

- facial keypoints **extraction** adds to preprocessing time, but the computational cost difference during model training is negligible
- **augmentation** techniques that produce useful variability in lower-dimensional data introduce excessive noise in high-dimensional representations => the model needs more samples to identify truly meaningful features

Bottom line: the limit is not in the computational cost, but in the amount of available data (**if** data augmentation helps on cleaner data!)

Next steps

Leveraging semi-annotated data:

- quality assessment:
  multi-stage filtering (from verifying segment boundaries and counting the number of people in frame to evaluating optical flow consistency),
  or training a classifier
- empirical evaluation of the usability threshold:
  completely removing low-quality samples
  or quality-based weighted learning?

Adjusting the pipeline:

- Sign2Gloss + Gloss2Text models
  (less sensitive to training data, according to Maia et al.)

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Thank you for your attention!

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